

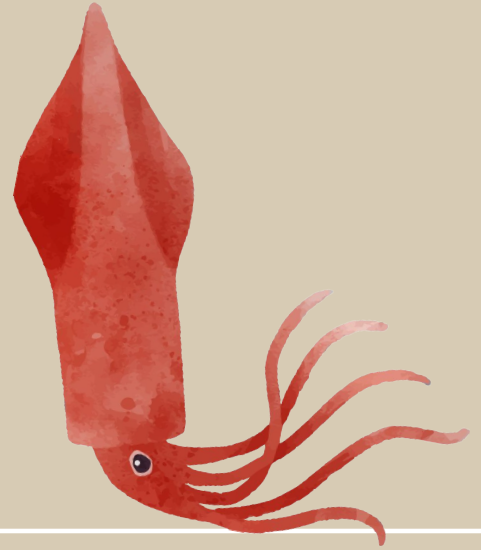
BIS 015L



Final Project

Ashlee Pilon and Maison Erridge

Quick Explanation of the Study



Fish were caught in Sarasota Bay, Florida and their tissues were examined for microplastics - tiny plastic particles under 5mm. The goal was to see if fish (prey) could be a source of plastic exposure for dolphins in the same bay.

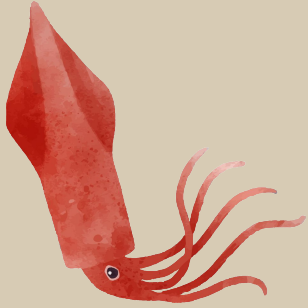
Overview of Study

Study Name: Microplastics in gastric samples from common bottlenose dolphins (*Tursiops truncatus*) residing in Sarasota Bay FL



“While some similarities in particle properties were shared between dolphins and fish, small sample sizes and incongruent findings for foams and TWP particles suggest that further investigation is warranted to understand the potential for trophic transfer.”

Sample Types



Fish Muscle

The flesh of the fish, plastics here means it got into the body tissue itself



Fish GI

The gastrointestinal tract (stomach + intestines), plastics here means the fish ate it



Particle Types



Fiber_single

Smooth, uniform surface
with a length that
exceeded the width



Fiber_bundle

20+ fibers tangled
together that can't
be separated



Film

Thin flat plastic sheet,
like plastic bag
fragments



Fragment_non TWP

Broken piece of harder
plastic, like a chunk off a
bottle or container



Fragment_TWP

Tire Wear Particles. Black
rubber fragments that come
off car tires.



Foam

Distorted when
handled, but goes back
to original shape



Fish Looked at in Study



Pigfish -
Orthopristis chrysoptera



Pinfish -
Lagodon rhomboides



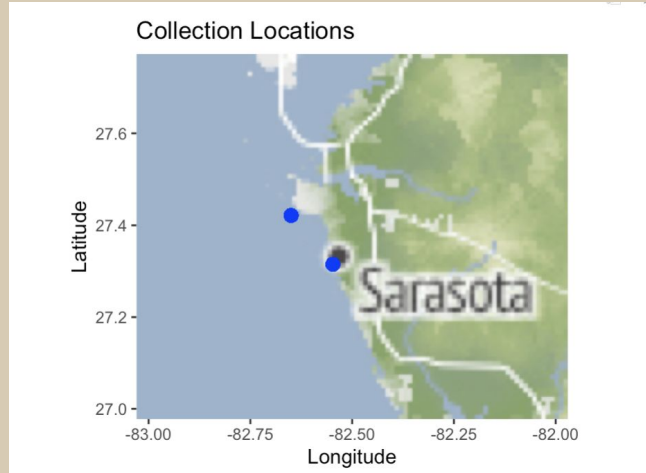
Gulf Toadfish -
Opsanus beta



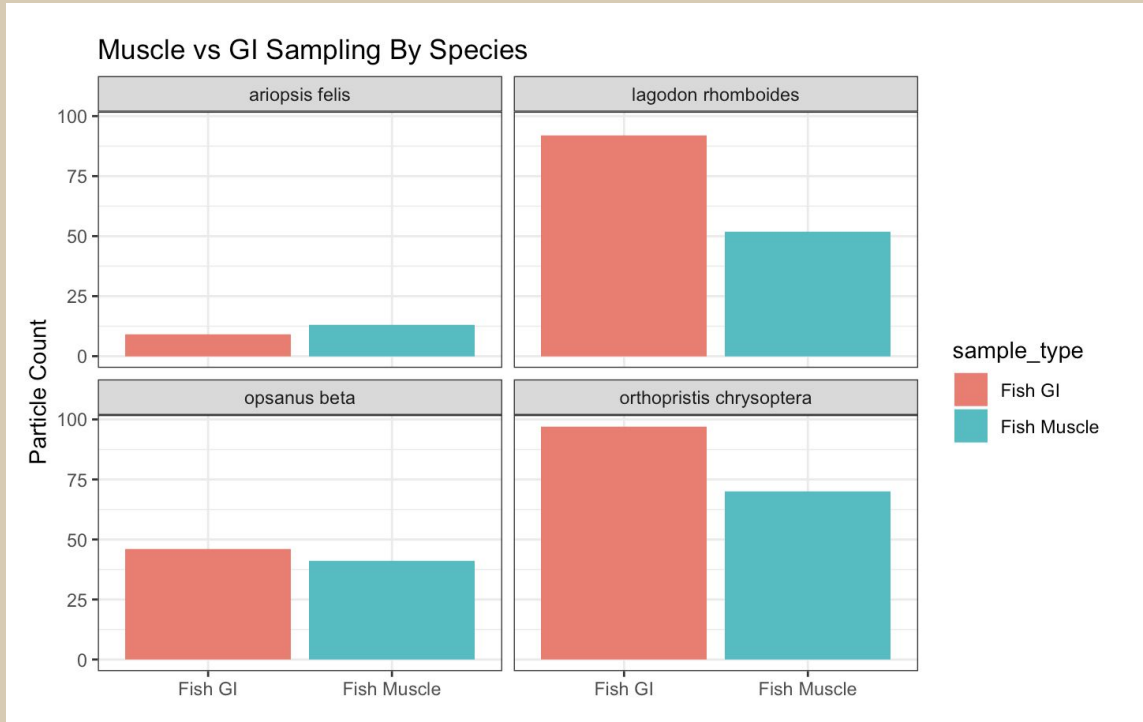
Hardhead Catfish -
Ariopsis felis

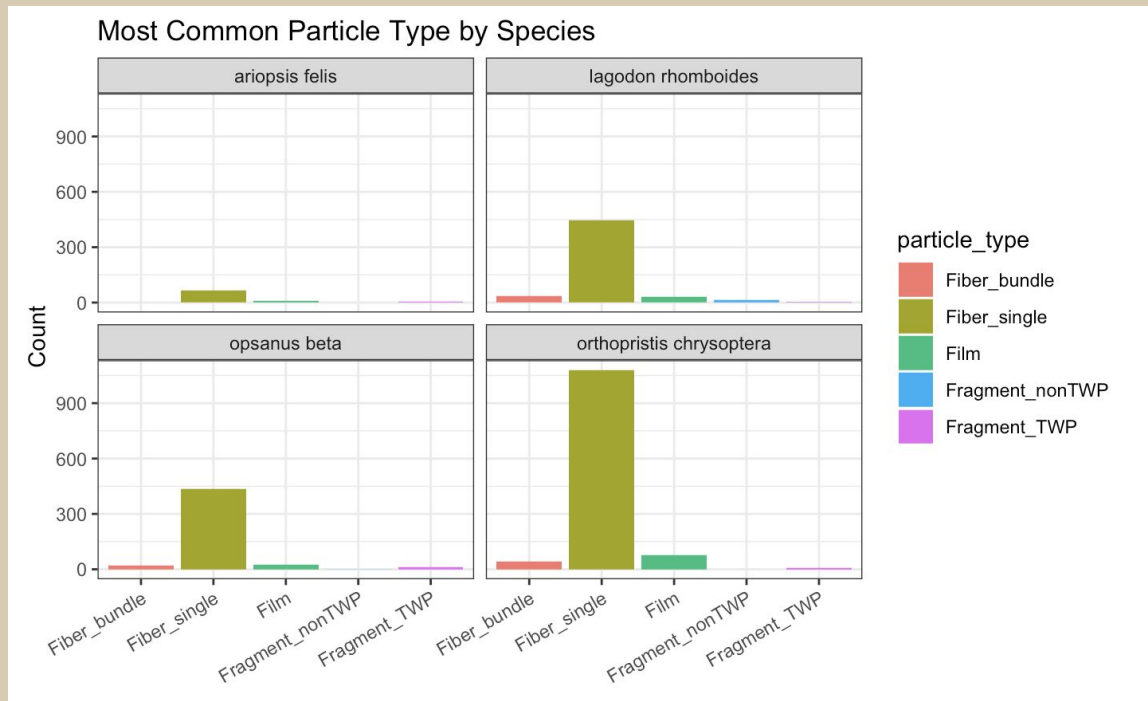


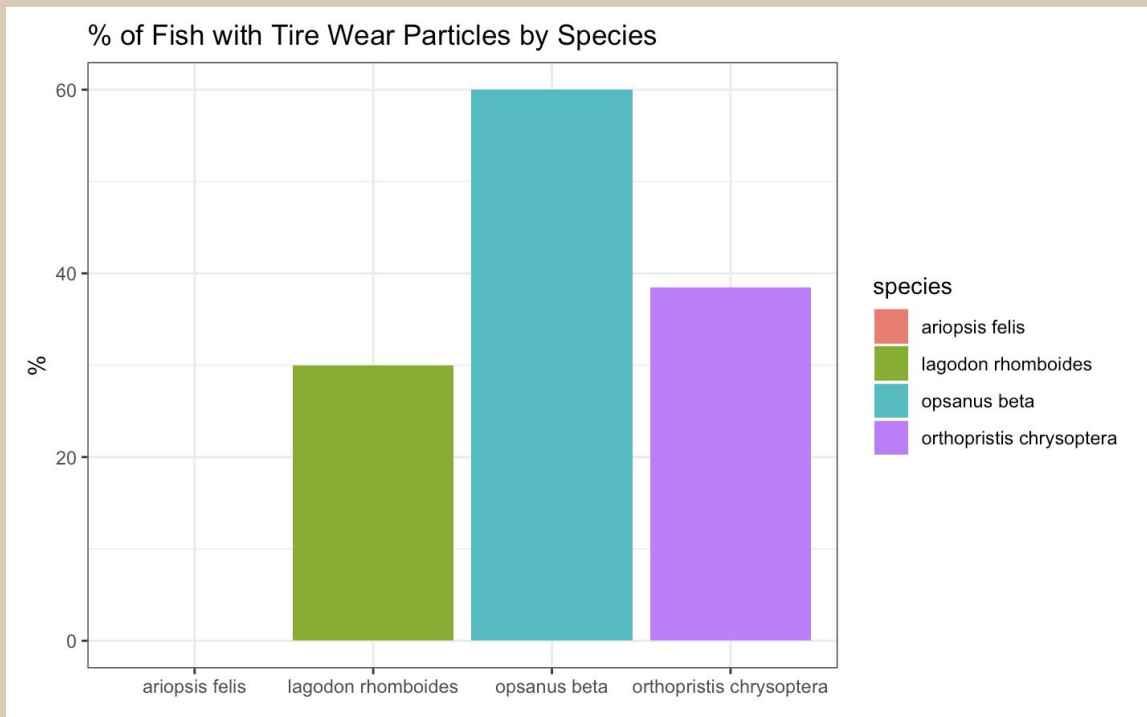
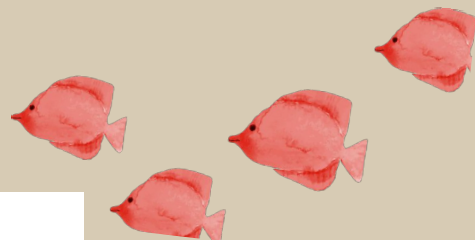
Where the Study Was Done

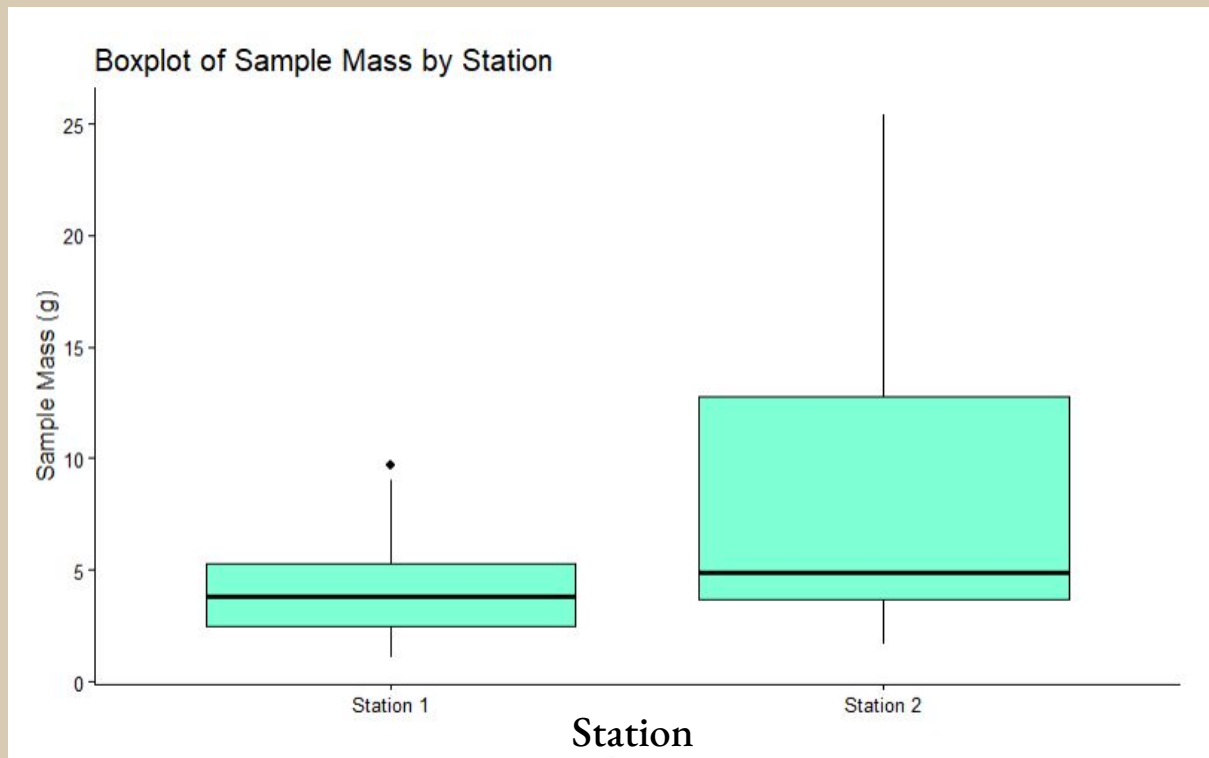


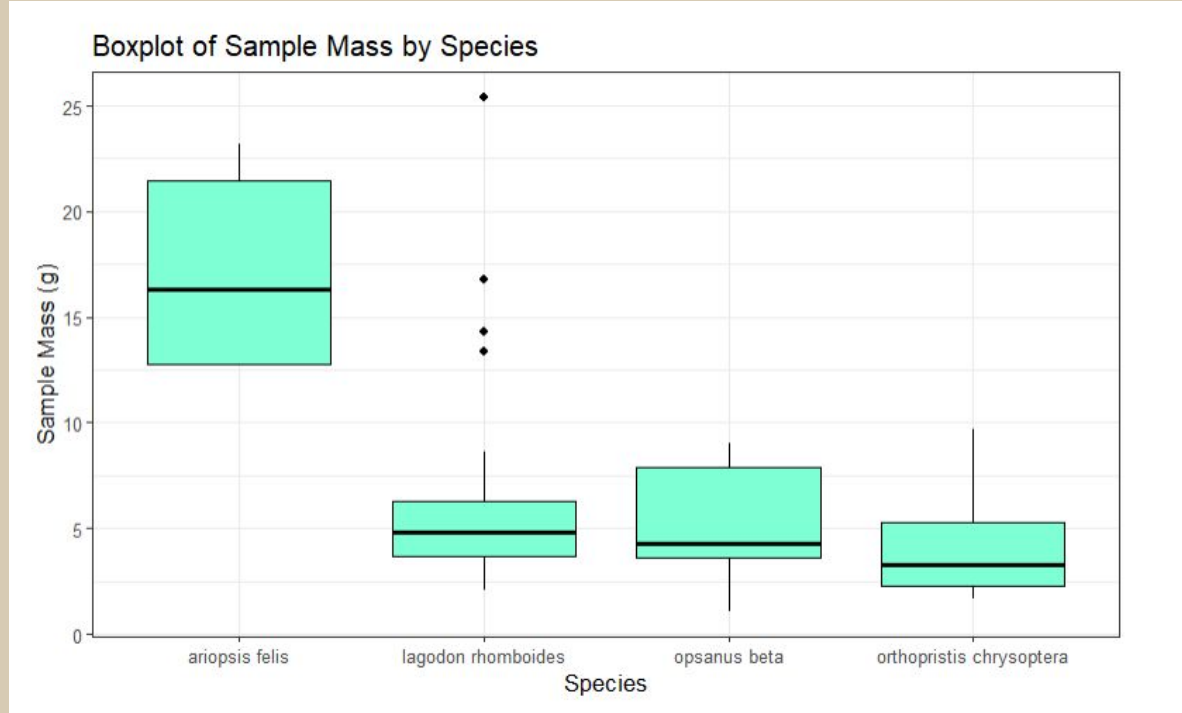
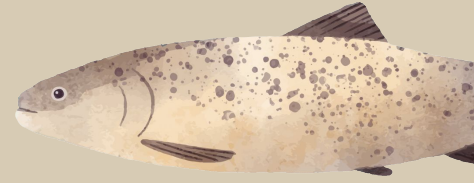
Graphs



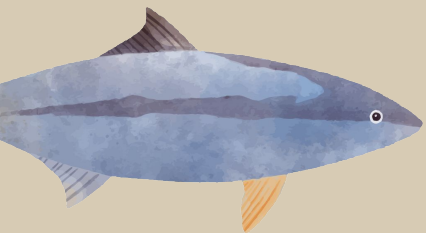








Shiny App



Takeaways From Data



- Microplastics were found in nearly every fish sampled, confirming that microplastic contamination is widespread in local fish populations.
 - Fiber_single particles dominated every species. We hypothesize that it originates from synthetic fabrics like polyester that shed during washing.
 - Tire wear particles showed up in 3 out of 4 species. 60% of toadfish contained tire rubber fragments.
 - Small sample sizes creates limited conclusions. With only 2 catfish and 5 toadfish, patterns in those species should be interpreted cautiously. A larger, more balanced sample across species would strengthen these findings.
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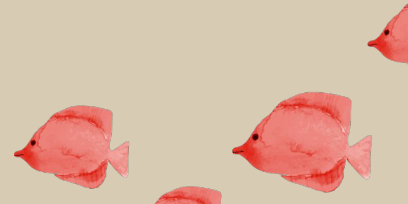
Future Research Questions

Bigger fish, more plastic?

Ariopsis felis had significantly larger sample masses than the other three species. Adding more catfish data would allow us to create a plastic to body mass ratio that would let us compare between species more effectively.

Does road runoff create microplastics?

TWP showed up in a majority of samples. Looking at samples of fish located near roads and highways could help assess if road runoff creates more microplastics.





Thank
You!

